

DS & AI

Machine Learning

Support Vector Machine

DPP 01

Q1 Consider the dataset with six datapoints: $\{(x_1,$

$y_1), (x_2, y_2), \dots, (x_6, y_6)\}$, where $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $x_2 =$

$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $x_3 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $x_4 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$, $x_5 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $x_6 =$

$\begin{bmatrix} -2 \\ -2 \end{bmatrix}$ and the labels are given by $y_1 = y_2 = y_5$

$= 1$, and $y_3 = y_4 = y_6 = -1$. A hard margin linear

support vector machine is trained on the above dataset.

Which ONE of the following sets is a possible set of support vectors?

- (A) $\{x_1, x_2, x_5\}$
- (B) $\{x_3, x_4, x_5\}$
- (C) $\{x_4, x_5\}$
- (D) $\{x_1, x_2, x_3, x_4\}$

Q2 In an SVM, what are support vectors?

- (A) Data points that lie closest to the decision boundary
- (B) Data points from all classes
- (C) Data points that are outliers
- (D) Data points with the highest dimensionality

Q3 In a linearly separable dataset, the optimal hyperplane in SVM is the one that:

- (A) Passes through the center of the data points
- (B) Maximizes the distance between support vectors
- (C) Minimizes the distance between support vectors
- (D) Creates the largest number of support vectors

Q4 What is the advantage of SVM over other classification algorithms?

- (A) It is immune to overfitting
- (B) It works only on linearly separable data
- (C) It always performs well with unbalanced datasets
- (D) It is effective in high-dimensional spaces and nonlinear data

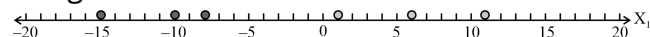
Q5 Pick the correct option from the below two statements:

S1: Consider a point which is correctly classified and is on the margin if we remove the point and the decision boundary changes then it is a support vector.

S2: Consider a point which is correctly classified and is on the margin if we remove the point and decision boundary doesn't change then it is not a support vector.

- (A) Only S1 is true
- (B) Only S2 is true
- (C) Both are true
- (D) Both are false

Q6 Find the decision boundary given by Hard-Margin SVM.



Q7 Every trained two-class hard-margin SVM model has at least one point of each class at a distance of exactly $\frac{1}{\|w\|}$ (the margin width) from the decision boundary. (Please enter 1 for True and 0 for false)

Q8 Suppose we have a soft-margin SVM for which $w = (-6, 3)$ and $b = 5$. Consider the example $(x = (2, 1), y = -1)$.

(I) How does the SVM classify x ?



(II) Does (x, y) satisfy the SVM constraint?
Mark the correct option for statement (I) and (II) respectively.

- (A) -1, Yes (B) +1, Yes
(C) -1, No (D) +1, No

Q9 Which of the following statements is true regarding the slack variable x in SVM classification?

- (A) $\xi > 0$ indicates that the point is classified correctly but lies inside the margin.
(B) $\xi = 0$ indicates that the point lies on the decision boundary.
(C) $\xi > 1$ indicates that the point is misclassified.
(D) $\xi < 0$ indicates that the point is classified correctly but outside the margin.

Q10 At the optimal solution of

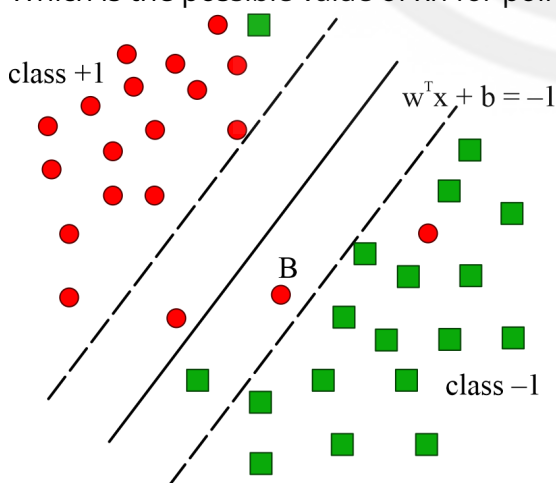
$$\min_{b, w, \xi} \frac{1}{2} w^T w + C \cdot \sum_{n=1}^N \xi_n$$

s.t. $y_n (w^T z_n + b) \geq 1 - \xi_n$ and $\xi_n \geq 0$ for all n ,

assume that $y_1 (w^T z_1 + b) = -10$. What is the corresponding ξ_1 ?

- (A) 1 (B) 11
(C) 9 (D) 31

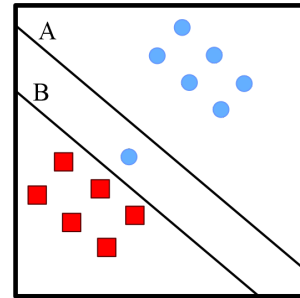
Q11 Which is the possible value of ξ_n for point B?



- (A) $\xi_n = 1$
(B) $\xi_n < 1$
(C) $1 < \xi_n < 2$
(D) $\xi_n > 2$

Q12 Which decision boundary do you prefer if little misclassifications are allowed?

(Please enter 1 for A and -1 for B)



Q13 What are support vectors:

- (A) The examples farthest from decision boundary.
(B) The only examples necessary to compute $f(x)$ in an SVM.
(C) The class centroids.
(D) All the examples that have a non-zero weight a_k in a SVM.

Q14

Suppose an SVM has $w = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $b = -1$.

What is the actual distance between the two planes defined by $w^T x + b = -1$ and $w^T x + b = 1$?

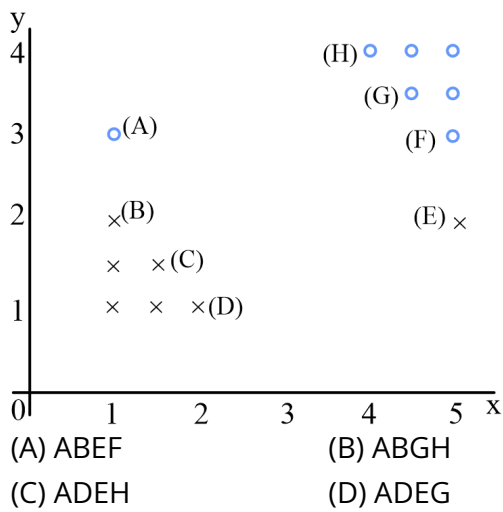
- (A) $\frac{1}{\sqrt{14}}$ (B) $\frac{2}{\sqrt{14}}$
(C) $\frac{\sqrt{11}}{2}$ (D) $\sqrt{11}$

Q15 If a hard-margin support vector machine tries to minimize $|w|^2$ subject to $y_i(X_i \cdot w + a) \geq 2$ instead, what will be the width of the slab (the point-free region bracketing the decision boundary)?

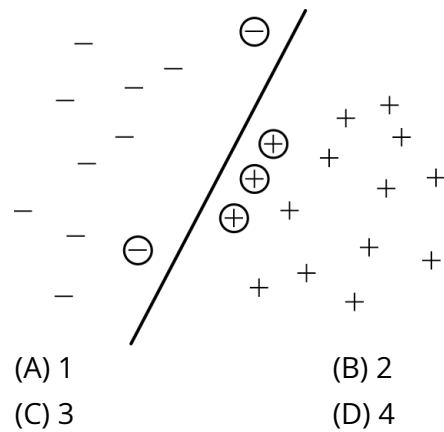
- (A) $\frac{1}{\|w\|}$ (B) $\frac{4}{\|w\|}$
(C) $\frac{2}{\|w\|}$ (D) $\frac{1}{2\|w\|}$

Q16 Select all points that are support vectors.





- Q17** Consider a binary classification problem in a 2-dimensional instance space $X = \mathbb{R}^2$. You are given a linearly separable training set. You run the hard-margin SVM algorithm and obtain the separating hyperplane below (support vectors are circled): What is the smallest number of data points that would have to be removed from the training set in order for the SVM solution to change?



- Q18** Which of the following statements is true regarding the slack variable $\xi(i)$ in SVM classification?
- (A) $\xi(i) > 0$ indicates that the point is classified correctly but lies inside the margin.
 - (B) $\xi(i) = 0$ indicates that the point lies on the decision boundary.
 - (C) $\xi(i) > 1$ indicates that the point is misclassified.
 - (D) $\xi(i) < 1$ indicates that the point is classified correctly but outside the margin.



Answer Key

Q1 D
Q2 A
Q3 B
Q4 C, D
Q5 A
Q6 0
Q7 1~1
Q8 A
Q9 C

Q10 B
Q11 C
Q12 0
Q13 A
Q14 B
Q15 B
Q16 A
Q17 A
Q18 C



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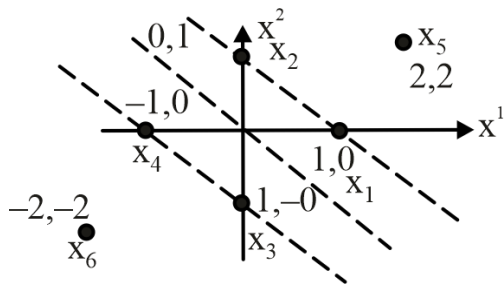


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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:



Q2 Text Solution:

• Data points that lie closest to the decision boundary-In **Support Vector Machines (SVM)**, **support vectors** are the **data points that lie closest to the decision boundary (hyperplane)**. These points are crucial because they define the margin of the classifier and help in determining the optimal decision boundary. Removing or changing these points can affect the position of the hyperplane.

Q3 Text Solution:

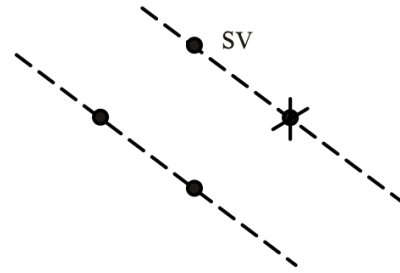
• Maximizes the distance between support vectors Max distance between SVM max margin.

Support Vector Machines (SVM), the **optimal hyperplane** is the one that **maximizes the margin**, which is the distance between the hyperplane and the closest support vectors from both classes. This ensures better generalization and robustness of the classifier.

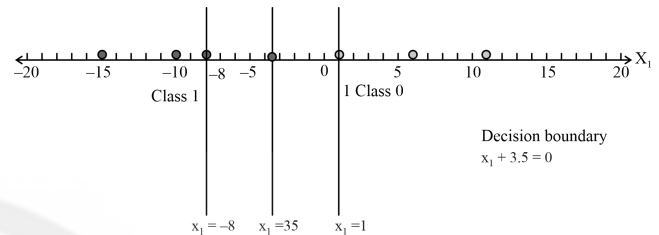
Q4 Text Solution:

- It always performs well with unbalanced datasets work well
- It is effective in high-dimensional spaces and nonlinear data

Q5 Text Solution:



Q6 Text Solution:



Q7 Text Solution:

Yes true

Q8 Text Solution:

$$\begin{aligned} & w_1 w_1 + w_2 + x_2 + b \\ & -6 \times 2 + 3 \times 1 + 5 \\ & -12 + 3 + 5 \\ & -4 \end{aligned}$$

• Point is below classifier class -1

Q9 Text Solution:

• $\varepsilon \geq 0$
 $0 < \varepsilon < 1 \rightarrow$ lie inside margin, classified correctly

Q10 Text Solution:

$$\min_{b, w, \xi} \frac{1}{2} w^T w + C \sum_{n=1}^N \xi_n$$

$$y_n(w^T z_n + b) \geq 1 - \xi_n, \quad \xi_n \geq 0, \quad \forall n$$

Now, we are given:

$$y_1(w^T z_1 + b) = -10$$

Using the constraint:

$$\xi_1 = \max(0, 1 - y_1(w^T z_1 + b))$$



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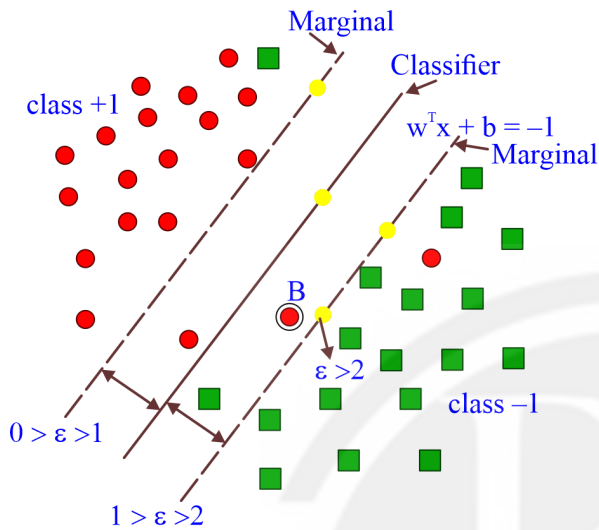
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Substituting $y_1(w^T z_1 + b) = -10$:

$$\xi_1 = 1 - (-10) = 1 + 10 = 11$$

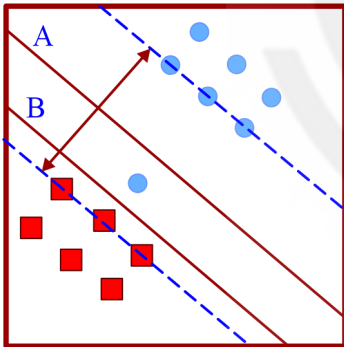
Q11 Text Solution:

- B point clam + 4
- B is wrongly classified
- $\epsilon > 1$



Q12 Text Solution:

- Classifier B zero error margin poor
- We choose A some misclassification allowed, margin will be large.



Q13 Text Solution:

A. The examples farthest from the decision boundary.

Support vectors are the data points that lie closest to the decision boundary (also called the margin) in a Support Vector Machine (SVM). These points are critical because they determine the position and orientation of the decision boundary.

Q14 Text Solution:

$$\frac{2}{\sqrt{1^2+2^2+3^2}} = \frac{2}{\sqrt{14}}$$

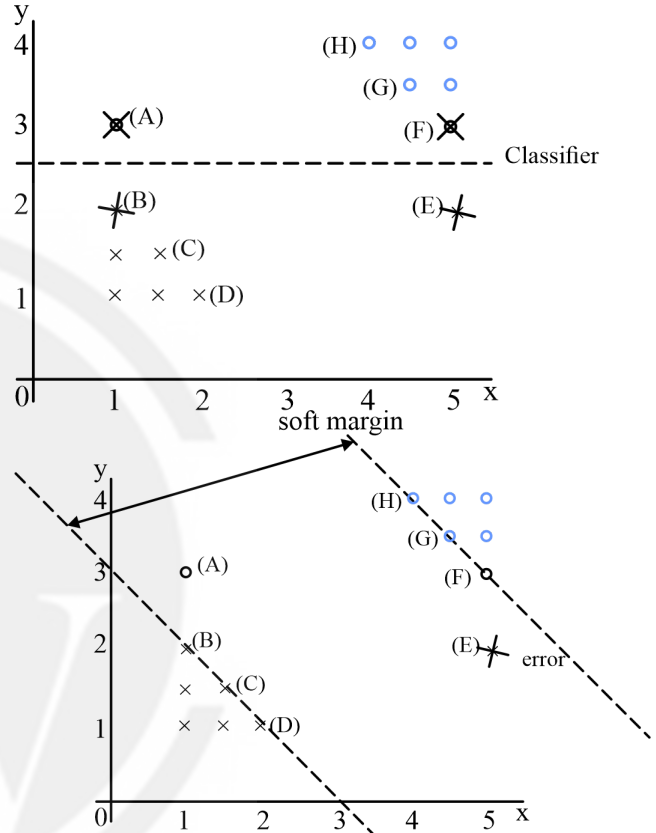
Q15 Text Solution:

$$wx_i + b = 2$$

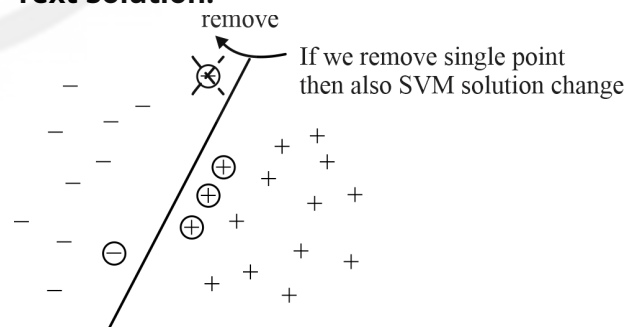
$$wx_i + b = -2$$

$$\text{distance} = \frac{4}{\|w\|}$$

Q16 Text Solution:



Q17 Text Solution:



Q18 Text Solution:

$0 < \xi < 1$: correct classify, but point is within margin

$\xi < 1$: misclassify



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